An Coimisiún um Rialáil Cumarsáide
Commission for Communications Regulation

Trinity College Dublin
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From Minutes to Megabytes:

Ireland's Telecom Transformation

Objective

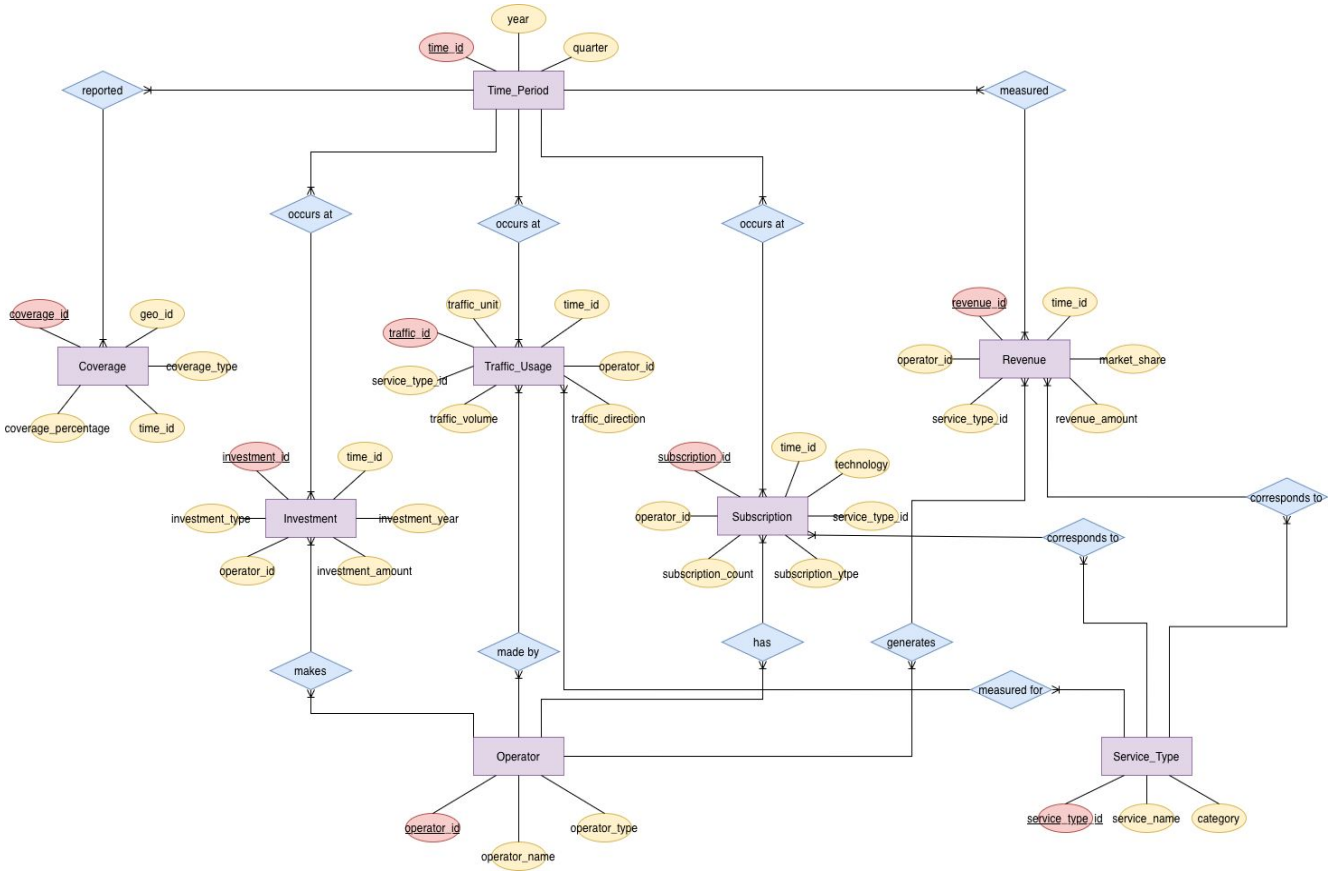
The objective of this project is to explore the evolving dynamic of Ireland's telecommunication sector over the period of 2015 to 2024. We are primarily focusing on the transition from the traditional fixed-line voices to mobile and broadband services. In spite of the technological innovations, the telecom industry providers continue to experience a challenge: the growth and the rapid expansion of the subscribes hasn't translated equally into revenue. The project aims to visualise and analyse the following mismatch, investigating the technological evolution, the shift in consumer behaviour and how the competitive markets have impacted and redefined the revenue streams. By understanding the gradual and progressive decline of fixed networks and the emergence of mobile and broadband, the project highlights the digital transition of Ireland, revealing how much telecom industry has evolved and what's on the horizon.

Project Overview

Dataset Selection

The data which was utilized to perform the comprehensive analysis of the telecom sector of Ireland was obtained from the **Irish Commission for Communications Regulation** (*Data Portal | Commission for Communications Regulation*, no date) , the official government body that regulates the electronic communications sector. ComReg's data portal supplies the quarterly datasets for telecom revenues, infrastructure coverage, and subscribers. Though the dataset covers the period of 2005–2025, only the data for 2015–2024 was considered in order to maintain completeness and dependability. This data is a mixture of behavioural, technological and financial perspectives with quarterly granularity, thus allowing in-depth trend analysis. The research is cantered on three main service pillars, Fixed Voice, Mobile and Broadband, specifically monitoring the number of subscriptions, traffic volumes, market shares, and revenue. Volume of traffic can be seen as a direct indicator of customer behaviour and usage intensity, hence the linking of subscriber growth to financial performance. Data on subscriptions and market shares indicates the market leaders, while revenue data indicates how the service usage has been converted into monetary value.

Entity-Relationship Diagram (ERD)



ERD illustrating the structure and relationships of Ireland's telecom data (Draw.io)

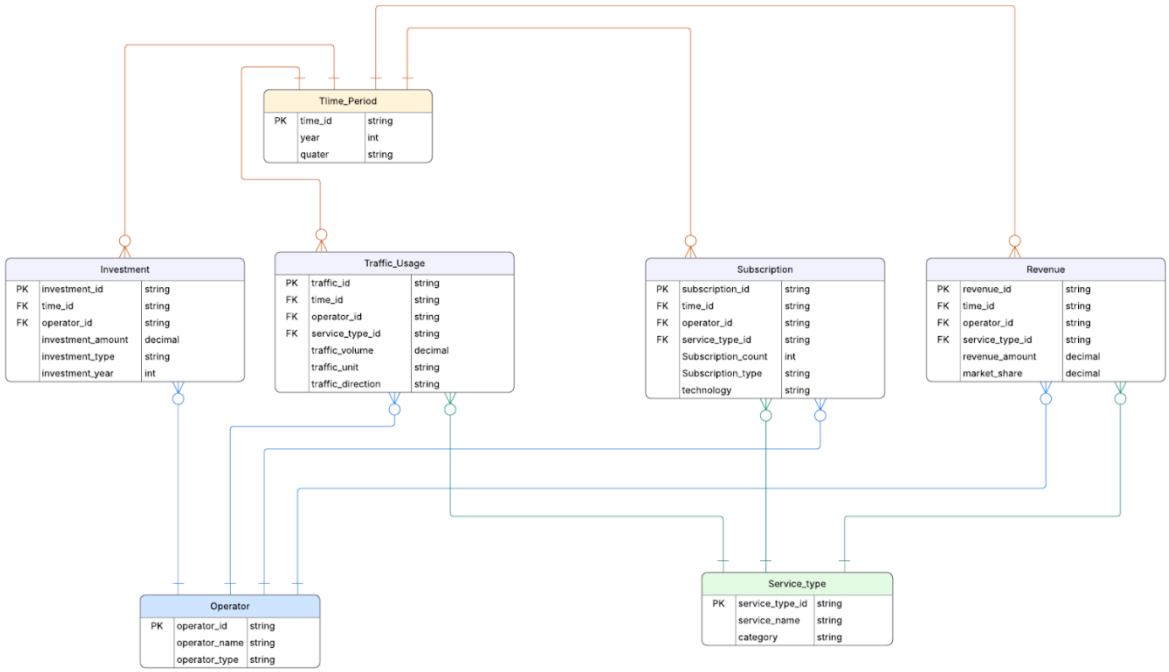
The Entity Relationship Diagram (ERD) illustrates how the different components of Ireland's telecom data connect to explain the industry trends between 2015 and 2024. The model includes entities such as: Operator, Subscription, Revenue, Traffic Usage, Service Type, Time period, Investment, Coverage in which each are identified by unique primary keys.

The Operator serves as the central actor, which is clearly responsible for revenue generation, traffic and subscription management, and finally making investment. Around it is the key analytical entities Subscription, Revenue and Traffic Usage, which clearly represents the three important perspectives (behavioural, financial and technological) as we mentioned before. Each of which connects to Service Type, which classifies the data into Fixed Voice, Mobile or Broadband, which allows to draw pretty clear comparison between traditional and modern communication services. Time period entity indicates when an investment occurs, coverage is reported, subscription happens and the other performance metrics are measured. Attribute

names in the ERD are standardized for clarity and may differ slightly from the original Excel column headers, but they accurately reflect the logical structure of the dataset.

Overall, the ERD provides a unified structure that links behaviour of subscribers, service adoption, and financial outcomes with operator activities and their time-based analysis. It finally provides a foundation for understanding Ireland's digital transformation.

Logical Database Model



Logical Data Model illustrating the relational structure of Ireland's telecom Data (Lucid Chart App)

The logical Data Model shows how the telecom datasets are connected and structured to support the analysis in Tableau. It translates the conceptual design of the ERD into much more detailed framework. The model includes four primary fact tables Subscriptions, Revenue, Investment, Traffic Usage and their corresponding dimension tables: Operator, Service Type, Time Period and Coverage. And each table can be uniquely identified by the primary key, for example Revenue has Revenue_id, this actually helps in maintaining the referential integrity while joining multiple tables.

The time period table connect with all performance-based entities, enabling the analysis by quarter and year and through one-to-many relationship, as each quarter can contain multiple

records across different metrics. Operator links everything together as a central point of reference for the telecom activities. Service Type helps in classifying whether the data is Fixed Voice, Mobile or Broadband, allowing direct comparisons among technologies. Investment records the financial input that actually drives the entire infrastructure growth. Whereas, Revenue, Subscription and Traffic Usage represent the main analytical measures.

While the data model defines how the entities are related, the actual excels used in Tableau was simplified and transformed for analysis. Some fields were merged, renamed and aggregated.

Data Transformation

The entire data preprocessing and transformation was performed in Microsoft Excel before then importing the data into Tableau. The original dataset ComReg contained an Excel workbook, with multiple excel sheets which represented different aspects of the Irish telecom industry between 2005 and 2025, recorded with a quarterly level granularity.

Step 1: Verifying and Reviewing Relevance

During the initial overview, we noticed numerous missing values and inconsistencies prior to 2015. Hence, we excluded the data from 2005 to 2014 for better data reliability. Similarly, we removed 2025 since it only covers Q1 and Q2, which would disturb total continuity. Therefore, the final analysis focuses on 2015 to 2024, where the data are complete and consistent.

Step 2: Data Segregation and Structuring

Each excel sheet containing data about Fixed Voice, Mobile and Broadband services consist of multiple types of information such as subscriber counts, traffic volumes etc To make the data easier to manage and analyse, these were separated into individual excel sheets based on their metric type, for example separate excel files for Subscription, Revenue, Investment, Traffic.

After the segregation each dataset was transposed (pivoted) to convert Quarter and Years into columnar form, so the structure can be compatible with Tableau's analytical format.

Step 3: Naming Convention Standardisation

All the datasets in the multiple excels were renamed with proper and consistent names and column headers, which immensely helps in maintaining uniformity across all the metrics and service types.

Step 4: Handling Missing and Incomplete Data

During validation, certain gaps were identified within the 2015-2024 range. Missing data in these gaps were handled carefully to maintain accuracy.

- In case of technologies introduced later, for example 5G, prior to Q2 2020 there was missing data as Ireland was newly introduced to 5G and the initial trail actually began in mid of 2020 and adoption was much slower due to COVID-19.
- For other minor gaps, mean imputation was applied to preserve the smooth trend without distorting the actual pattern.

This approach actually helped in ensure that every time-series remained realistic.

Step 5: Key Creation and Data Integration

The implementation of a distinctive Time_ID as a foreign key across all the datasets made it possible to have coherence and integration in Tableau. Besides, the Year and Quarter columns were added for clarity in each file together with unique identifiers like Revenue_ID for ensuring the uniqueness of the records. It is worth noting that although no independent time table was made, Time_ID acted as the primary linking key that connected Subscription, Traffic, Revenue, and Investment datasets.

Step 6: Final Validation and Quality Check

Before the datasets were imported into Tableau, a thorough check was conducted for missing values and duplicates. Filtering and grouping were facilitated by the addition of metric and segment columns. This meticulous operation not only resulted in a clean dataset but also a well-structured and standardized one, thus guaranteeing the accurate and consistent visualization of Ireland's telecom industry story in Tableau.

Tableau Visualization

https://public.tableau.com/app/profile/ahmad.kamran1294/viz/Group6_DVM_17610684133390/MarketDynamics

Dashboard 1

Irish Telecom Overview (2015–2024)

The initial dashboard named “Irish Telecom Overview (2015-2024)” gives a comprehensive overview of the performance of Ireland’s telecommunications market for the last ten years.

Three KPIS, from the period from 2015 to 2024 witnessed an increase of 2.8% in Mobile Subscriptions, while the rise in Broadband Subscribers was 2.92%, thus, pointing out the strong digital uptake. On the other hand, Fixed Voice Subscribers fell by 23.21%, which was a clear indication of the gradual extinction of traditional voice communication.

The chart of *Mobile ARPU* reveals an uninterrupted drop of average revenue per user which is a nuancing of the market competition and the boosting of the user base at the same time with the individual spending being the lowest. Mobile Revenue, on the other hand, has been quite stable implying that the loss of ARPU was balanced with the total users' growth.

The graph of *Broadband Usage vs Subscribers* makes it clear that the increase in the number of subscribers was rather slow whereas the broadband traffic grew very significantly, thus, it could be inferred that per user data consumption went up. This is a clear indication of Ireland’s increasing dependence on high-speed digital connectivity.

Lastly, the *Technology-wise Broadband Trends* chart illustrates the transitioning from DSL to FTTP and cable technologies which has been encouraged by the fibre rollout initiatives. There is a clear shrinkage in the DSL lines and a boom in FTTP lines over the decade- the encouragement for this change comes from the higher speeds that can be provided through FTTP and the backing of the government’s digital infrastructure plans (*Government approves ambitious Digital Connectivity Strategy for Ireland*, no date).

In brief, the dashboard provides a sneak peek into the major structural changes in the Irish telecom market, a shift from fixed-line dependency to an advanced and connected economy powered by mobile and broadband services.

Dashboard 2

Technology Shifts: The Mobile – Broadband Transition

This dashboard showcases the quickly changing Irish telecom landscape, which is moving away from traditional fixed-line and mobile services and going towards broadband and next-generation technologies such as 4G and 5G. The chart comparing Broadband vs Mobile Data Usage illustrates a drastic reduction in 2G and 3G users, the prevalence of 4G, and the gradual rise of 5G since 2020, which are all indications of Ireland's continuous transition to fast, data-oriented, and high-quality connectivity.

The growth of broadband subscriptions is another indicator of the increasing reliance of Irish consumers on data connectivity, both mobile and fixed.

The chart of *Voice to Data: The Landline Decline* demonstrates that although the fixed voice revenue has been steadily declining, mobile and broadband revenues have been gaining, thus attesting to the transition in the telecom sector from traditional telephony to data-driven services.

In the same way, *Fixed Voice vs Mobile Traffic* shows that there has been a huge drop in landline usage and the opposite is true for mobile usage. This clearly reflects a change in customers' preferences. Lastly, Mobile vs Broadband Revenue indicates that by 2024, revenue from broadband will surpass that from mobile data, a significant event which signifies the shift of broadband to be the primary revenue source in the Irish telecom market.

Dashboard 3

Market Dynamics

The dashboard overall presents the competition between mobile and broadband services in Ireland. Along with that, it also highlights the country's transition to data-driven connectivity through higher investments in broadband infrastructure.

The *Mobile Market Share* chart shows subscription growth by operator and their respective shares. The market is stable: Vodafone leads slightly, followed by Three and Eir. While smaller players also stay stable over the years.

The *Broadband Market Share* picture demonstrates a clear development of market consolidation over time, so that now Eir, Virgin Mobile and Vodafone are the kings of the broadband world. The smaller ones, such as Sky and Others, still have considerable shares, but Eir still holds the top spot by 2024.

Fixed vs Mobile Investment, at the bottom right demonstrates the contrast in capital investment trends in Fixed and Mobile infrastructures. The investment in broadband networks saw a massive increase during the decade, reaching its highest point around 2023. The investment has been very fruitful considering the sharp increase in the increase in revenue from broadband. The investment in mobile infrastructure is steady but secondary. This is indicative of a balanced infrastructure policy growing both mobile strength and fixed broadband access concurrently.

Conclusion

The visualizations from 2015 to 2024 drastically point out the telecoms of Ireland industry has gone through a radical change. The usage of fixed-line voice services by the customers has always been decreasing, meanwhile, mobile and broadband connections have turned out to be the two main things that not only kept the market alive but also brought in further investments. Broadband traffic was growing and the use of fibre-based technologies was increasing quickly with the backing of the government's digital infrastructure plans, while mobile operators were stepping up to 4G/5G level- meeting the increased data demands. A falling ARPU was not able to hurt mobile revenues since the number of subscribers was increasing. The competition in the market and the rising investments in broadband point to the sector's acknowledgement of the importance that broadband holds. In short, the visualizations together tell a story of the digital transition of Ireland, a decade characterized by the switch from the old telephony system to high-speed, data-oriented connectivity which is going to determine the communications of the future.

References

Data Portal | Commission for Communications Regulation (no date). Available at: https://www.comreg.ie/industry/electronic-communications/data-portal/downloads/ (Accessed: October 21, 2025).

Government approves ambitious Digital Connectivity Strategy for Ireland (no date) *gov.ie*. Available at: https://www.gov.ie/en/department-of-climate-energy-and-the-environment/press-releases/government-approves-ambitious-digital-connectivity-strategy-for-ireland/ (Accessed: October 21, 2025).